

From Relation-Specific Heads to Shared Interventions: Contextual Entrainment in Language Models

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Abstract

Large language models rely on prompt context to perform in-context learning, follow formats, and adapt to local evidence. This ability, however, also raises a robustness concern: models may reuse irrelevant tokens from earlier context when answering a later query. Prior work identifies this behavior as *contextual entrainment* and finds attention heads whose ablation reduces it, arguing that these heads are largely task- or relation-specific. But are the mechanisms behind contextual entrainment truly isolated by relation, or do relation-conditioned searches recover reusable components shared across relations? Using factual-distractor relation-completion prompts across 19 lexical-semantic subtypes, we identify attention heads whose ablation reduces entrainment and test how the resulting head sets transfer recovery ability across exact relations, semantic families, and mixed relation distributions. Same-family transfer exceeds different-family transfer, family residuals weaken once globally shared heads are removed, and subtype-specific searches repeatedly rediscover many global heads. Clean-completion and context-use probes further suggest that this shared intervention can reduce harmful copying while preserving ordinary relation completion and simple in-context abilities. Our findings provide a more precise perspective on contextual entrainment and its mitigation: harmful context following is not best addressed by searching for isolated relation-specific circuits, but by using heterogeneous attack prompts to reveal reusable intervention targets whose effects are shaped by semantic relation structure.

1 Introduction

Large language models exhibit a wide range of context-dependent behaviors: they follow instructions, infer task formats from demonstrations, maintain discourse entities, and answer questions based on prompt-local evidence (Brown et al., 2020; Olsson et al., 2022). These abilities, often grouped

under in-context learning, entail a sensitivity to previous tokens.

This matters in settings where models must reason over long or heterogeneous inputs, including retrieval-augmented generation, long-context question answering, and tool-augmented interaction (Shi et al., 2023; Yoran et al., 2024; Cuconasu et al., 2024; Wu et al., 2024; Cheng et al., 2025). In such settings, irrelevant context is not always random noise: it can provide structured cues that models incorporate into later predictions. Cheng et al. (2025) make this concern concrete through hallucinations in irrelevant-contexts, showing that models can combine abstract class cues with features drawn from the query or context, producing systematic but unreliable answers. Our setting studies a token-level version of this problem: the distractor sentence is factually true for another subject, but its answer token becomes misleading evidence for a later, structurally parallel query.

This token-level route is closely related to *contextual entrainment*. Following Niu et al. (2025), we use this term for cases where tokens that appear earlier in the prompt become more likely to be generated later, even when they do not answer the current query. For example:

The capital of Japan is Tokyo. The
capital of China is

can increase the probability of the distractor token Tokyo, even though the correct answer is Beijing. The first sentence is true, and the repeated relation frame makes the earlier city name immediately available at the second answer position, where it competes with the correct continuation.

Recent mechanistic work has begun to localize this behavior inside transformer models. Niu et al. (2025) identify attention heads whose ablation reduces contextual entrainment and argue that the recovered head sets are largely task- or relation-specific. Related work on knowledge conflict, how-

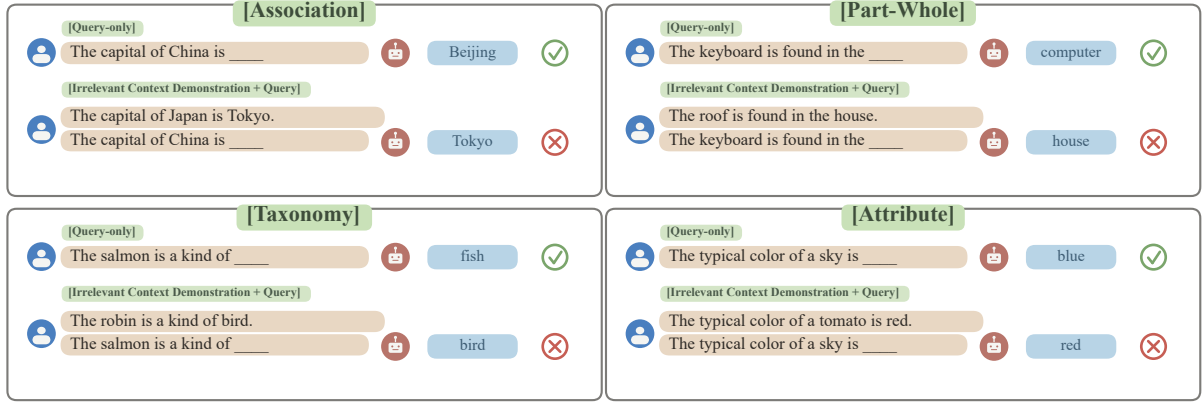


Figure 1: Factual-distractor examples across four lexical-semantic families.

ever, shows that head-level interventions can sometimes generalize across relations and prompt formats (Jin et al., 2024). This distinction matters for both interpretation and mitigation. If entrainment heads are isolated relation-specific mechanisms, then contextual entrainment is organized into many narrow circuits, and mitigating it may require searching separately for each relation. If relation-conditioned searches instead recover overlapping shared components, then apparent task specificity may reflect how a reusable context-following mechanism is selected under different data slices. This leaves an open question: when a head set is discovered from a single relation, is it an isolated relation-specific mechanism, or a mixture of components with different scopes?

This question gives a head-level complement to Cheng et al. (2025): if irrelevant context systematically shapes factual predictions, we ask whether the interventions that reduce this sensitivity are relation-specific or shared across relations. We hypothesize that relation-conditioned searches recover mixtures of shared and relation-sensitive components. Rather than treating each relation-specific search result as a separate module, we ask whether these head sets contain components that operate at different levels of generality. We refer to this as a mixed-circuit account of contextual entrainment. This account makes different predictions from two simpler alternatives. If entrainment-related heads are purely relation-specific, a head set found on one subtype should transfer little beyond that subtype. If they are fully general, semantic relatedness should not affect transfer. The mixed-circuit account predicts the middle case: strongest transfer to the same subtype, weaker but positive transfer within the same semantic family, and weak-

est transfer across unrelated families.

We test this hypothesis on a controlled lexical-semantic benchmark with 19 relation subtypes grouped into Association, Attribute, Part-Whole, and Taxonomy. As illustrated in Figure 1, each clean relation-completion prompt is paired with an attacked prompt that prepends a true but irrelevant parallel fact from the same subtype. The answer token in this prepended fact serves as the distractor: it appears in the immediate context, but it is not the correct continuation for the later query. This lets us isolate prompt-local token reuse without introducing an explicit false statement about the queried subject. We then search for entrainment-related heads and evaluate how the resulting interventions transfer across same-subtype, same-family, different-family, and mixed-relation conditions.

Contributions. We make three contributions. First, we show that entrainment-related interventions transfer across lexical-semantic relations in a semantically graded way, challenging a purely relation-specific interpretation. Second, we identify a rediscovered global core: mixed-relation search recovers a reusable component that overlaps with subtype-specific searches and better preserves clean external relation completion than subtype-specific search. Third, we validate a mixed-circuit decomposition in which global, family-residual, and subtype-residual components combine approximately additively at the level of causal intervention effects.

2 Related Work

Distraction and contextual entrainment. LLMs are highly sensitive to prompt context, and

prior work shows that irrelevant or misleading evidence can substantially alter model outputs, especially in retrieval-augmented generation and long-context question answering (Shi et al., 2023; Yoran et al., 2024; Cuconasu et al., 2024; Wu et al., 2024). Cheng et al. (2025) further show that irrelevant contextual cues can induce structured hallucinations through class-based misgeneralization. Niu et al. (2025) formalize *contextual entrainment*: tokens that appear earlier in the prompt are more likely to be generated later, even when they are random or semantically irrelevant. They identify attention heads whose ablation reduces entrainment and interpret the recovered head sets as largely task- or relation-specific.

Attention heads as context mechanisms. Mechanistic interpretability has shown that attention heads can implement identifiable computations, including induction-like copying, name moving, and task-specific circuits (Olsson et al., 2022; Wang et al., 2023). Induction heads are especially relevant here because they implement a form of pattern completion: continuations from earlier contexts are reused at later matching positions (Olsson et al., 2022). Related work on knowledge conflict further suggests that attention heads can mediate competition between parametric memory and external context, and that head-level interventions can generalize across relations and prompt formats (Jin et al., 2024). Together, these findings make it plausible that factual-distractor prompts share a reusable context-following component even when their surface relations differ.

Circuit discovery by differentiable masking. A growing line of work uses activation patching, pruning, and differentiable masks to identify causally relevant components in language models. Yu et al. (2025) use Gumbel-Sigmoid masking for gradient-based joint computation-graph pruning in language-model circuit discovery, building on continuous relaxations of discrete variables and sparse stochastic gates (Maddison et al., 2017; Jang et al., 2017; Louizos et al., 2018). Niu et al. (2025) apply a related Gumbel-Sigmoid head-search framework to contextual entrainment. These methods make it possible to move beyond descriptive transfer patterns and ask whether head sets discovered from individual relations reflect isolated mechanisms or repeatedly recovered shared structure.

Shared and distributed mechanisms. Broader mechanistic work argues that model mechanisms can be distributed, overlapping, and partly redundant rather than cleanly modular (Elhage et al., 2022; McGrath et al., 2023). This matters for head search: low overlap between recovered head sets is not, by itself, evidence for isolated task-specific mechanisms. At the same time, work on relation representations suggests that factual relations are systematically organized in model states (Hernandez et al., 2024). Together, these perspectives motivate our question: whether relation-conditioned causal searches recover narrow local effects, or broader shared structure with relation-sensitive modulation.

3 Methodology

We first define the attacked relation-completion task and intervention, then describe how head sets are searched and evaluated, before introducing the benchmark and transfer conditions used to test our hypothesis. All main experiments use LLaMA-3.1-8B (AI@Meta, 2024). Interventions are performed by zero-ablating selected attention head outputs during the forward pass, while leaving all model parameters unchanged.

3.1 Task Definition

We define contextual entrainment as an attacked relation-completion task with four entities and answers: a *query subject* s_q , its *gold answer* y_{gold} , a different *distractor subject* s_d , and its *distractor answer* y_{dist} . Using the country–capital example in Figure 1, China is the query subject, Beijing is the gold answer, Japan is the distractor subject, and Tokyo is the distractor answer. We use this terminology throughout the paper.

Each example is built from a relation template $T(\cdot)$. The clean prompt queries the answer for the query subject alone, while the attacked prompt prepends a structurally parallel factual statement about the distractor subject:

$$x_{\text{clean}} = T(s_q), \quad x_{\text{attack}} = T(s_d) + y_{\text{dist}} + T(s_q).$$

The attacked prompt places a concrete distractor answer in the local context of the later query, where it competes with the gold answer at the shared answer position.

We use *factual distractors* rather than counterfactual context. The prepended statement is true for another subject but irrelevant to the current query.

This design isolates prompt-local token reuse from explicit memory–context conflict, making the attack weaker but cleaner. Our goal is to test whether a true earlier answer for one subject interferes with the correct continuation for another.

3.2 Gumbel-Sigmoid Head Search

To identify candidate anti-entrainment circuits, we run a differentiable head-search procedure in the spirit of Niu et al. (2025). Each attention head receives a learnable binary mask m_i , where $m_i = 1$ zero-ablates the head output. We optimize these masks with a Gumbel-Sigmoid straight-through relaxation (Maddison et al., 2017; Jang et al., 2017; Louizos et al., 2018), which allows gradient-based search over discrete head subsets. We then use the learned mask scores to define ranked top- K interventions for later analyses.

For attacked prompts, the objective maximizes recovery of the gold-over-distractor margin while encouraging sparsity:

$$\mathcal{L} = -\mathbb{E}_{x \in \mathcal{B}} \left[M_{\theta}^{\text{abl}}(x) - M_{\theta}(x) \right] + \frac{\lambda}{|\mathcal{H}|} \sum_i \sigma(\alpha_i).$$

After search, the learned mask scores define a head ranking, from which we export top- K interventions. We use $K = 80$, selected by a held-out sweep balancing attack recovery and clean-prompt harm. Additional optimization details are in Appendix H.

3.3 Evaluation Metrics

We evaluate interventions by how much they restore the model’s preference for the gold answer over the distractor answer. For a prompt x , let $\ell(y \mid x)$ be the model logit for token y at the answer position. We measure this preference with the margin

$$M(x) = \ell(y_{\text{gold}} \mid x) - \ell(y_{\text{dist}} \mid x).$$

The factual distractor is harmful when it lowers this margin, so we define attack drop as

$$D = M(x_{\text{clean}}) - M(x_{\text{attack}}).$$

Because context following is generally useful, we evaluate recovery only on *valid attacked cases*: the clean prompt must favor the gold answer and the distractor must reduce the margin by at least $D \geq 0.5$. This isolates cases where prompt-local context induces a measurable factual-distractor effect.

For a head set S , the raw causal effect of zero ablation is

$$\Delta_S = M_S(x_{\text{attack}}) - M(x_{\text{attack}}),$$

where M_S is the margin after ablating S . Positive Δ_S means the intervention shifts the model away from the distractor and toward the gold answer. Since target subtypes differ in attack strength, we also report normalized recovery,

$$R_S = \frac{\Delta_S}{D},$$

where $R_S = 1$ indicates full recovery of the attack-induced margin loss (e.g., if ablating S improves the attacked margin by $\Delta = 0.5$ on both *country–capital* (France–Paris) with $D = 0.5$ and *part–whole* (wheel–car) with $D = 2$, then R is 1 vs. 0.25). Values above 1 indicate over-recovery: the intervention increases the attacked margin by more than the margin loss induced by the distractor. We therefore interpret such values as intervention effects rather than bounded percentages. Similar over-recovery effects are reported by Niu et al. (2025).

For transfer analyses, we further remove target-specific recoverability by centering normalized recovery within each target:

$$\tilde{R}_{s,t} = R_{s,t} - \frac{1}{|\mathcal{S}|} \sum_{s' \in \mathcal{S}} R_{s',t}.$$

This asks whether a source circuit performs better than the average source circuit on the same target (e.g., on target *country–capital* (France–Paris), if most sources cluster around $R \approx 0.6$ but a same-family source reaches $R = 0.8$, then $\tilde{R} \approx +0.2$ captures above-average transfer rather than target ease). We use $\tilde{R}$ for same-family versus different-family comparisons, while also reporting raw margin changes Δ .

3.4 Lexical-Semantic Benchmark

Having defined the task, intervention, and evaluation metrics in general terms, we now instantiate them on a controlled lexical-semantic benchmark with 19 relation subtypes grouped into four families: Association, Attribute, Part–Whole, and Taxonomy. Taxonomic, part–whole, and attribute relations are standard organizing relations in lexical semantics and WordNet-style lexical databases (Cruse, 1986; Miller, 1995; Fellbaum, 1998); Association covers non-taxonomic factual and functional mappings common in relation-completion

Family	Sub.	Pairs	Valid test	Rate	$\bar{D}$
Association	4	163	441/480	91.9	3.92
Attribute	5	244	293/600	48.8	2.69
Part-Whole	5	224	492/600	82.0	4.77
Taxonomy	5	227	496/600	82.7	3.60
Total	19	858	1722/2280	75.5	3.86

Table 1: Benchmark statistics by semantic family after tokenizer and valid-attack filtering. **Valid test** uses the main filter $D \geq 0.5$; $\bar{D}$ is the mean attack drop.

benchmarks (Petroni et al., 2019; Hernandez et al., 2024). Representative examples and statistics are shown in Figure 1 and Table 1, respectively. More details are reported in Appendix A.

3.5 Transfer and Circuit Conditions

A circuit searched on source subtype s is evaluated on target subtype t . We group source–target pairs as *same-subtype*, *same-family*, or *different-family* transfer. To analyze recovered circuits, we compare *subtype-specific*, *family-shared*, *global-shared*, and *family-residual* head sets, where family residuals exclude the global-shared top- K heads from the corresponding family ranking. We also measure head overlap and define frequency-global heads by rediscovery frequency across subtype-specific searches.

4 Results

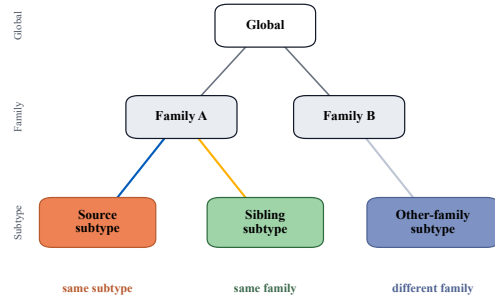
The results test whether relation-conditioned head searches identify isolated relation-specific circuits or mixed circuits with shared components. At first glance, searches on different lexical-semantic subtypes return different top- K heads, and these differences matter for transfer. The key result, however, is that this surface-level task dependence masks a shared structure: relation-conditioned searches repeatedly include a global component, with family- and subtype-level variation layered around it.

We organize this pattern using the mixed-circuit hypothesis. For a subtype of relation r , a set of heads recovered by relation-specific search can be functionally viewed as

$$C_r \approx G + F_{\text{fam}(r)} + S_r + \epsilon,$$

where G is a reusable global context-following component, $F_{\text{fam}(r)}$ is semantic-family modulation, S_r is subtype-specific structure, and ϵ captures template, optimizer, or factual-completion idiosyncrasies. This decomposition is not intended to be a

(a) Transfer conditions in the semantic hierarchy



(b) Transfer is graded by semantic relatedness

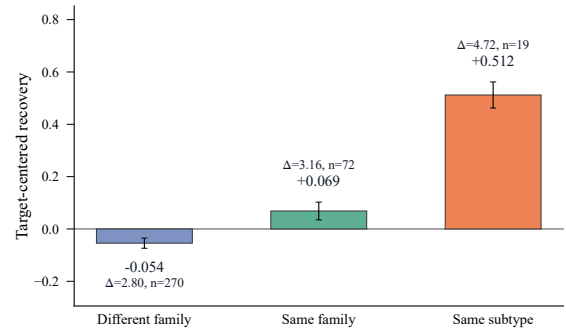


Figure 2: Transfer is graded by semantic relatedness. The top panel illustrates the transfer conditions from a source circuit to target relations. The bottom panel reports target-centered recovery, which controls for target-specific recoverability.

strict linear factorization of head effects; rather, it provides a guide to interpret transfer, residual, and overlap patterns.

We first verify that the recovered head sets have genuine causal effects on factual-distractor entrainment and that transfer is graded by semantic relatedness (Figure 2). We then use transfer, residualization, head overlap, additive intervention tests, and clean-capability probes to distinguish isolated task-specific circuits from the mixed-circuit account.

4.1 Transfer Is Graded by Semantic Relatedness

Figure 2 summarizes this structure: the transfer matrix falls between two extremes—isolated relation-specific circuits and fully universal heads.

Figure 2 shows a graded pattern: different-family transfer is weakest, same-family transfer is stronger, and same-subtype transfer is strongest. After target-centering, different-family transfer is below the target average (-0.054), same-family

Circuit	Assoc.	Attr.	P-W	Tax.	Avg.
Global shared	1.502	1.801	1.028	1.531	1.464
Family shared	1.486	1.904	0.925	0.943	1.306
Family residual	0.881	1.243	0.699	0.997	0.959
Subtype-specific	1.606	1.619	1.069	1.361	1.404

Table 2: Shared-circuit decomposition. Values are normalized recovery. Family residuals exclude global-shared top- K heads from the family ranking.

transfer is above it (+0.069), and same-subtype transfer is much larger (+0.512). This hierarchy remains significant under clustered bootstrap, cell-level tests, and family-label permutation controls; full statistics are reported in Appendix B.

Relation-specific searches therefore do not recover isolated subtype modules, but neither do they recover a fully relation-agnostic circuit. Semantic family and exact subtype both modulate transfer.

4.2 Family Effects Are Built on Global Shared Heads

The graded transfer pattern suggests that semantic relatedness modulates transfer, but it does not by itself determine whether this structure reflects family-specific modules or variations around a shared mechanism. We therefore compare four circuit constructions: circuits searched on a balanced mixture of all active subtypes (*global-shared*), circuits searched only within the target family (*family-shared*), circuits searched on the exact target subtype (*subtype-specific*), and family rankings after removing global-shared top- K heads (*family residual*).

Table 2 supports the mixed-circuit account. The global-shared circuit is the strongest aggregate intervention, with average recovery 1.464, while family-shared circuits remain effective but weaker on average (1.306). Subtype-specific circuits are also strong (1.404), indicating that exact relation structure still contributes. Thus, family-conditioned search is useful, but it is not the dominant source of recovery.

The residual comparison provides more direct evidence. Once globally shared heads are removed from the family ranking, average family recovery drops from 1.306 to 0.959. The residual remains positive, so family structure is not absent. However, the drop shows that much of the apparent family-shared effect is carried by heads that also belong to the global circuit. In the decomposition, G carries the largest recoverable component, while

Head set	$K = 20$	$K = 40$	$K = 80$
Global search	0.207	0.724	1.464
Frequency-defined global	0.364	0.639	1.350
Frequency $\cap$ global	0.361	0.788	1.431
Subtype-specific	0.301	0.654	1.404
Family-shared	0.204	0.551	1.306
Rare heads	0.004	0.014	0.082

Table 3: Frequency-defined global heads recover most of the global circuit’s effect, while rarely rediscovered heads do not. Values are normalized recovery.

$F_{\text{fam}(r)}$ and S_r account for the remaining semantic variation in transfer. Thus, same-family transfer is real, but much of this family structure is organized around a reusable global core rather than independent family modules.

4.3 Relation-Specific Searches Rediscover the Global Core

A strong global circuit could still be an artifact of mixed-relation search. If so, it would not need to appear inside relation-specific circuits. The mixed-circuit account predicts the opposite: the same global heads should be repeatedly rediscovered when searching on individual subtypes.

This prediction is borne out in the overlap statistics. Global-shared circuits overlap with family-shared circuits by 40.25 heads on average, compared with 6.25 expected by random overlap. They overlap with subtype-specific circuits by 32.84 heads on average, and 75 of the 80 global-shared heads appear in at least one subtype-specific top- K circuit. The global circuit is therefore not separate from relation-specific circuits; it is largely made of heads that narrower searches also select.

Rediscovery frequency is itself predictive of causal effect. If we rank heads only by how often they appear in subtype-specific top- K circuits, the resulting frequency-defined global heads recover much of the global-search circuit. At $K = 80$, they reach recovery 1.350, close to the global-search recovery 1.464, while rarely rediscovered heads recover only 0.082 (Table 3). This provides head-level evidence that G is already embedded in the subtype circuits C_r .

4.4 The Mixed-Circuit Decomposition Predicts Intervention Effects

The mixed-circuit account treats a relation-conditioned head set as a combination of a global component, a family-level residual, and a subtype-level residual.

For each target subtype, we construct disjoint head sets corresponding to the global component G , the family-residual component F , and the subtype-residual component S , and evaluate interventions that include different subsets of these components. We then fit a regression over raw margin recovery:

$$\Delta = \beta_0 + \beta_G I_G + \beta_F I_F + \beta_S I_S + \epsilon,$$

where I_G , I_F , and I_S indicate whether the corresponding component is ablated.

The fitted coefficients support the mixed-circuit decomposition. The global component has the largest effect ($\beta_G = 2.936$), while family and subtype residuals also contribute positively ($\beta_F = 1.009$, $\beta_S = 1.159$). The additive model explains substantial variance in raw recovery ($R^2 = 0.572$); adding all interaction terms barely improves fit ($R^2 = 0.578$), and additive and full-interaction predictions are nearly identical ($r = 0.995$). Thus, the components are not independent modules, but their causal effects are well approximated by an additive decomposition.

A fixed-budget diagnostic supports the same conclusion. An 80-head mixture combining G_{40} , F_{20} , and S_{20} achieves the strongest recovery among matched 80-head interventions, outperforming pure global, family-only, and subtype-specific alternatives. This indicates that G carries the dominant recoverable effect, while $F_{\text{fam}(r)}$ and S_r add smaller but useful residual contributions. Details in Appendix F.

4.5 Global Circuits Preserve Clean Relation Completion

A final test asks whether the global circuit reduces entrainment by broadly disrupting relation-completion behavior rather than by selectively counteracting the distractor. If ablating it only damaged general relation completion, then its recovery effect would be less meaningful. We therefore run a clean capability evaluation on external LRE relations, where prompts contain no distractor context.

Figure 3 summarizes the clean-capability probe. The global circuit preserves external relation completion closely, and family-shared circuits remain similarly close to the base model on average. By contrast, subtype-specific circuits cause larger collateral degradation across the 19 subtype-specific interventions. The figure uses the country-capital subtype as an example of a narrow subtype-specific intervention; the full subtype-specific breakdown is reported in Appendix D.

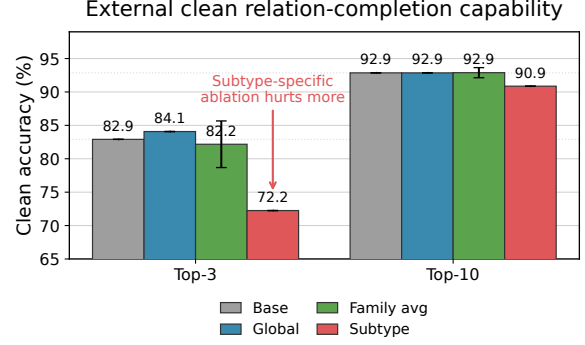


Figure 3: External LRE clean relation-completion capability after ablation.

This pattern is consistent with the idea that mixed-relation search is not merely broader, but cleaner. One plausible explanation is that single-relation search can entangle distractor suppression with ordinary factual-completion machinery for that relation. Heterogeneous search instead favors heads whose ablation consistently reduces factual-distractor entrainment across relation environments while better preserving clean relation completion.

5 Robustness and Reproducibility

The core empirical pattern is stable under several additional checks: transfer remains semantically graded, and the shared global component remains useful across alternative analyses.

Controls: Uniformly random and same-layer random head controls are near zero or negative, indicating that recovery is not a generic consequence of ablating many heads or ablating heads from sensitive layers (Appendix B). **Thresholds, templates, and seeds:** Same-family transfer remains stronger than different-family transfer across valid-attack thresholds from $D \geq 0.25$ to $D \geq 1.0$. Cross-template recovery remains positive, and an eight-subtype three-seed matrix preserves the same qualitative hierarchy. **Model replication:** We replicate the main transfer experiments on Mistral-7B (Jiang et al., 2023) and Qwen2.5-7B (Yang et al., 2024) under the same evaluation protocol. **Additional probes:** A context-use probe suggests that global ablation does not notably damage ordinary in-context learning ability, as reflected in simple few-shot spelling correction, translation, and in-context mapping tasks. Appendix E further shows that the global effect is not specific to zero ablation via clean-to-attacked activation replacement.

Detailed results are reported in Appendices B–E.

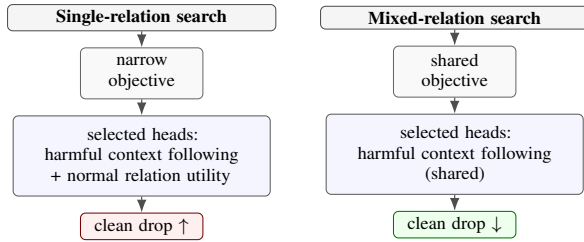


Figure 4: Selection pressure comparison between single-relation and mixed-relation search.

6 Discussion

From task-specific heads to mixed circuits. Our results refine the task-specific interpretation of contextual entrainment heads proposed by Niu et al. (2025). Consistent with prior work, searches on different relation tasks do not return identical head sets. This variability is not surprising: transformer computations are often distributed and redundant, with multiple components able to support overlapping functions or compensate for one another under intervention (Elhage et al., 2022; McGrath et al., 2023). The key question is therefore not whether different tasks select exactly the same heads, but whether the selected heads are functionally isolated. Our results suggest that they are not. Relation-conditioned searches repeatedly contain a global component that transfers across semantic families, while family and subtype structure modulate this shared component.

Why a global core is significant. Contextual entrainment is behaviorally general: models often reuse prompt-local structure across tasks and relations (Olsson et al., 2022; Niu et al., 2025). However, this behavioral generality does not by itself imply a shared circuit-level intervention. The same vulnerability could have been implemented by largely separate relation-specific head sets, in which case a circuit found for one relation would transfer little beyond that relation. Our results show instead that relation-conditioned searches repeatedly recover a strong relation-general component. Semantic family and subtype structure still shape the effect, but mainly by modulating this shared component rather than replacing it with separate modular circuits. Thus, the significance of the global core is mechanistic: apparent relation specificity partly reflects different uses of a reusable context-following substrate.

Why mixed search can reduce side effects. Single-relation search entangles two components:

heads that reduce distractor copying and heads involved in normal relation completion. This explains why subtype-specific circuits are strong but have larger side effects. Heterogeneous search changes the selection pressure: a head is favored only if its ablation consistently reduces distractor entrainment across different relations. The resulting global circuit better preserves ordinary completion while targeting harmful context following.

Implication for RAG and factual context use.

The global component also suggests a practical direction for settings with heterogeneous factual context. In retrieval-augmented generation, multi-document question answering, dialogue history, and tool-augmented agent traces, the model may see true facts that are useful locally but irrelevant to the current query. A mitigation method therefore cannot simply suppress context use, since factual grounding is the purpose of these systems. Prior relation-specific interventions may be difficult to deploy in such settings, where a single context can mix many relation types and factual roles. In contrast, mixed-relation search offers a more targeted alternative: it identifies a reusable intervention that reduces copying from factual distractors while largely preserving ordinary relation completion. This extends the RAG motivation of contextual entrainment from diagnosis to intervention, suggesting that diverse factual-distractor examples can help localize shared targets for reducing harmful context copying.

7 Conclusion

By analyzing entrainment across lexical-semantic relations, we show that context following in LLMs is neither fully relation-specific nor fully relation-general. Head-level intervention and transfer analyses reveal mixed circuits: a reusable global component supports broad context following, while semantic-family and subtype structure shape how this component is used in particular relations. These findings challenge the view that entrainment-related heads form isolated relation-specific modules. Apparent task specificity instead emerges from overlapping shared and relation-sensitive components. For mitigation, this suggests starting from heterogeneous attack data to identify broad reusable interventions, while using relation-specific searches for fine-grained specialization.

8 Limitations

Several limitations remain. Although we replicate the core transfer pattern on Mistral-7B and Qwen2.5-7B, the detailed decomposition, overlap, and frequency analyses are conducted primarily on LLaMA-3.1-8B. Our benchmark is also controlled: prompts use relation-completion templates, and answers are restricted to single tokens so that gold-versus-distractor margins are directly comparable. This design improves interpretability, but it leaves open whether the same mixed-circuit structure appears in more natural long-context or retrieval settings. We did not conduct our experiments using larger LMs due to resource limitations.

Our conclusions are also conditional on valid attacked cases. This filtering is appropriate for studying harmful entrainment, since recovery is meaningful only when the factual distractor actually weakens the gold-over-distractor margin. However, it does not characterize all uses of context following, including cases where context helps the model answer correctly. Finally, head ablation identifies causally useful components, but it does not fully specify the internal algorithm by which the model resolves competition between the gold answer and the distractor. Future work should combine head-level interventions with activation patching or representation-level analyses, and should evaluate stability over more seeds and broader prompt distributions. We will release the code, prompts, and analysis scripts upon acceptance to support reproducibility and follow-up work.

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A Benchmark Details

Table 8 lists the 19 relation subtypes used in the benchmark. Each subtype contains factual (x, y) pairs and two relation-specific templates. We filter examples so that both the gold answer and the distractor answer are single tokens under the model tokenizer.

Tables 8 and 9 provide the subtype and template details.

Definition of Association. We use ASSOCIATION as an operational cover family for relation subtypes that are not taxonomic, meronymic, or attributive, but are linked by conventional thematic, functional, institutional, or usage-based knowledge (e.g., country–capital, profession–workplace, sport–equipment, and task–tool). This follows computational lexical-semantics work emphasizing that many useful word-pair relations are broader, non-classical relational links rather than WordNet-style taxonomic or part–whole relations (Morris and Hirst, 2004; Turney, 2006; Jurgens et al., 2012). We therefore treat ASSOCIATION as a heterogeneous contrast family, not as a single classical semantic primitive.

Overlapping family boundaries. The family labels are operational groupings rather than mutually exclusive lexical-semantic primitives. Some relation types naturally sit near the boundary between families. Material relations are a clear example: they can be viewed as substance meronymy when the material is treated as a constituent of an object, but also as attribute-value relations when the material is treated as a property of the object. We therefore retain material-based subtypes in both the Part–Whole and Attribute families instead of manually removing these boundary cases. This choice makes the family-transfer test more conservative: overlap between families should blur, rather than sharpen, same-family versus different-family distinctions. The fact that we still observe graded same-subtype, same-family, and different-family transfer under this operational grouping supports the robustness of the semantic-structure result.

B Additional Transfer Statistics

Valid-attack filtering. Our main experiments define valid attacked cases as examples where the clean prompt favors the gold answer and the factual distractor reduces the gold-over-distractor margin by at least $D \geq 0.5$. This retains 1,722 of

2,280 LLaMA-3.1-8B test attacks across 19 relation subtypes. Table 4 shows that the same-family advantage is stable across nearby threshold choices.

Threshold	Valid tests	Same-family	Diff.-family	Raw adv.	Centered adv.	p
$D \geq 0.25$	1776	0.997	0.871	0.125	0.128	0.0022
$D \geq 0.50$	1722	0.956	0.838	0.118	0.123	0.0021
$D \geq 0.75$	1660	0.926	0.812	0.113	0.118	0.0021
$D \geq 1.00$	1590	0.885	0.779	0.106	0.112	0.0024

Table 4: Valid-attack threshold sensitivity. Same-family transfer remains stronger than different-family transfer across all tested attack-drop thresholds. Same-family and different-family columns report non-target-centered normalized recovery. The centered advantage is computed after target-centering; p is the cell-level Welch test for the centered same-family versus different-family comparison.

Dependency-aware inference. For the same-family advantage test, we use target-centered recovery and additionally run a two-way clustered bootstrap over source and target subtypes. For each bootstrap sample, we resample source and target subtypes with replacement, reconstruct the corresponding transfer cells, recompute target-centered recovery, and then recompute the same-family minus different-family advantage.

Statistic	Observed	Median	95% CI	p
Same-family advantage	0.123	0.115	[0.003, 0.236]	0.024

Table 5: Two-way clustered bootstrap over source and target subtypes. The statistic is the target-centered same-family minus different-family advantage.

Random baselines. Table 6 reports the full random and same-layer random controls. Each cell reports non-target-centered normalized recovery, with raw attacked-margin change in parentheses.

Method	Diff.-family	Same-family	Same-subtype
Gumbel	0.693 (2.711)	1.081 (3.737)	1.407 (5.418)
Random	-0.167 (-0.695)	-0.089 (-0.395)	-0.123 (-0.458)
Same-layer random	-0.202 (-0.659)	-0.117 (-0.349)	-0.213 (-0.818)

Table 6: Random and same-layer random controls.

Transfer condition	Target-centered recovery	Raw Δ
Different-family	-0.054	2.80
Same-family	+0.069	3.16
Same-subtype	+0.512	4.72

Table 7: Target-centered transfer by semantic relatedness. Target-centering removes target-specific recoverability before comparing source circuits.

Graded transfer detail

Family	Subtype	Example pair	Representative template
Association	country–capital	France → Paris	The capital of France is Paris.
Association	profession–workplace	doctor → hospital	The workplace of a doctor is a hospital.
Association	sport–equipment	tennis → racket	The equipment used in tennis is a racket.
Association	task–tool	writing → pen	The tool used for writing is a pen.
Attribute	color	banana → yellow	The typical color of a banana is yellow.
Attribute	shape	coin → round	The typical shape of a coin is round.
Attribute	material	window → glass	The typical material of a window is glass.
Attribute	temperature	ice → cold	The typical temperature of ice is cold.
Attribute	size	elephant → large	The typical size of an elephant is large.
Part–Whole	component–object	wheel → car	The wheel is a component of the car.
Part–Whole	member–collection	player → team	The player is a member of the team.
Part–Whole	portion–whole	slice → cake	The slice is a portion of the cake.
Part–Whole	material–object	table → wood	The table is made of wood.
Part–Whole	body-part–body	finger → hand	The finger is part of the hand.
Taxonomy	animal	sparrow → bird	The sparrow is a kind of bird.
Taxonomy	plant	rose → flower	The rose is classified as a flower.
Taxonomy	artifact	hammer → tool	The hammer is a type of tool.
Taxonomy	food	apple → fruit	The apple is a type of fruit.
Taxonomy	vehicle	sedan → car	The sedan is a kind of car.

Table 8: Relation subtypes and representative templates in the lexical-semantic benchmark.

Subtype	Template 1	Template 2
country–capital	The capital of {x} is {y}.	The capital city of {x} is {y}.
profession–workplace	The workplace of a {x} is a {y}.	A {x} typically works in a {y}.
sport–equipment	The equipment used in {x} is a {y}.	People use a {y} in {x}.
task–tool	The tool used for {x} is a {y}.	People use a {y} for {x}.
color	The typical color of a {x} is {y}.	The color commonly associated with a {x} is {y}.
shape	The typical shape of a {x} is {y}.	The shape commonly associated with a {x} is {y}.
material	The typical material of a {x} is {y}.	A {x} is typically made of {y}.
temperature	The typical temperature of {x} is {y}.	A {x} is typically {y} in temperature.
size	The typical size of a {x} is {y}.	A {x} is typically {y} in size.
component–object	The {x} is a component of the {y}.	The {y} contains a component called a {x}.
member–collection	The {x} is a member of the {y}.	The {y} includes the member {x}.
portion–whole	The {x} is a portion of the {y}.	The {y} contains a portion called a {x}.
material–object	The {x} is made of {y}.	The material of the {x} is {y}.
body-part–body	The {x} is part of the {y}.	The {y} contains the body part {x}.
animal	The {x} is a kind of {y}.	The {x} is classified as a {y}.
plant	The {x} is a kind of {y}.	The {x} is classified as a {y}.
artifact	The {x} is a type of {y}.	The {x} is classified as a {y}.
food	The {x} is a type of {y}.	The {x} is classified as a {y}.
vehicle	The {x} is a kind of {y}.	The {x} is classified as a {y}.

Table 9: Prompt templates used for each relation subtype.

C Robustness and Cross-Model Replication

This appendix reports robustness checks for template variation, seed stability, and cross-model replication.

Template robustness. Cross-template recovery remains positive but weaker than same-template recovery: Association drops from 1.671 to 1.488, Attribute from 1.855 to 1.610, Part-Whole from 0.801 to 0.690, and Taxonomy from 1.721 to 0.968.

Seed stability. In an eight-subtype mini transfer matrix over three seeds, same-family transfer remains positive (0.071, 0.276, 0.259), same-subtype transfer is larger (0.510, 0.726, 0.631), and different-family transfer remains negative (−0.097, −0.167, −0.148).

Cross-model replication. Table 10 reports non-target-centered transfer recovery for all three models under valid-attack filtering.

Model	Diff.-family	Same-family	Same-subtype
LLaMA-3.1-8B	0.838	0.956	1.404
Mistral-7B	0.674	0.835	1.229
Qwen2.5-7B	2.210	2.717	3.566

Table 10: Cross-model transfer summary. Values are non-target-centered normalized recovery under valid-attack filtering.

D External Clean and Context-Use Probes

Circuit	Top-3	Top-10
Base	82.90	92.86
Global shared	84.07	92.86
Family-shared avg.	82.17	92.89
Subtype-specific avg.	78.50	91.72
Subtype-specific: country-capital	72.25	90.87

Table 11: External LRE clean capability. Global-shared ablation preserves clean relation completion, while subtype-specific circuits cause larger collateral degradation on average.

External LRE clean capability. We evaluate relation-completion prompts without distractor context. The global mixed-relation circuit preserves clean relation-completion accuracy, while the subtype-specific circuits cause larger top-3 degradation.

Relation	Top-3			Top-10		
	Base	Global	Subtype	Base	Global	Subtype
fruit outside color	40.0	56.7	13.3	86.7	83.3	86.7
landmark in country	90.0	91.5	88.0	95.5	96.0	95.5
landmark on continent	84.5	84.0	76.5	97.0	97.5	97.0
product by company	85.5	86.0	84.0	93.5	93.0	91.0
star constellation name	85.6	87.8	41.7	89.2	89.9	80.6
task done by tool	68.8	71.9	68.8	90.6	90.6	90.6
task person type	80.0	80.0	80.0	85.0	85.0	85.0
work location	57.6	51.5	60.6	75.8	72.7	75.8
Overall	82.9	84.1	72.2	92.9	92.9	90.9

Table 12: External LRE clean capability test (full per-relation breakdown; corresponds to Figure 3 in the main text). Base denotes no ablation; Global denotes the mixed-relation global circuit; Subtype denotes the country-capital subtype-specific circuit.

Context-use preservation. We also evaluate simple context-use tasks, including few-shot spelling correction, few-shot translation, and synthetic in-context mapping. Global ablation leaves the spelling and translation probes unchanged at 100% top-10 accuracy across all tested shot counts. On the harder synthetic mapping probes, performance is more variable, with the largest degradation on nonce-label mapping. Overall, top-1 accuracy is unchanged, top-3 accuracy drops from 39.71 to 38.60, and top-10 accuracy drops from 66.91 to 61.03 (Table 13).

Circuit	<i>n</i>	<i>K</i>	Top-1	Top-3	Top-10	ΔTop-3	ΔTop-10
Base	272	0	31.62	39.71	66.91	0.00	0.00
Global shared	272	80	31.62	38.60	61.03	1.10	5.88

Table 13: Expanded context-use preservation probe. Accuracies and drops are in percentage points; positive drops mean worse than the base model.

E Activation Replacement Robustness

To test whether the global component is specific to zero ablation, we run a clean-to-attacked activation replacement intervention. For each valid attacked prompt, we first run the corresponding clean prompt and cache the selected heads’ final-position outputs. We then run the attacked prompt while replacing those same head outputs at the answer position with their clean-prompt activations. This intervention is less destructive than zero ablation and tests whether the selected heads carry attack-sensitive contextual signal.

Figure 5 reports the results on 100 valid attacked cases, balanced across the four semantic families. Global-shared heads recover 60.4% of the attack-induced margin loss under activation replacement,

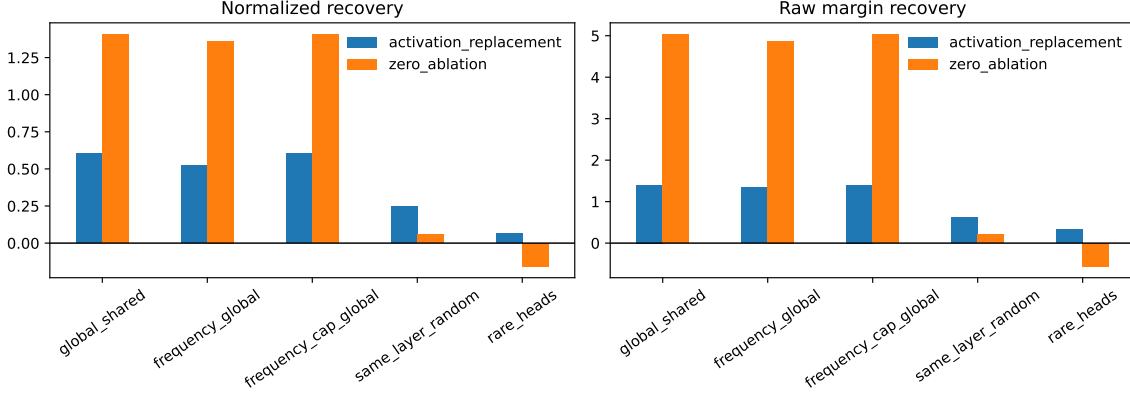


Figure 5: Activation replacement robustness. Clean-to-attacked activation replacement recovers substantial attacked margin for global and frequency-defined global heads, while rare heads remain weak. Zero ablation is stronger, but the replacement result shows that the global component is not merely a zero-ablation artifact.

with a raw margin increase of 1.40. Frequency-defined global heads show a similar effect, recovering 52.4% of the margin loss. By contrast, rare heads recover only 6.5%. The logit decomposition further distinguishes global heads from controls: global replacement both increases the gold logit and decreases the distractor logit, whereas rare and same-layer random heads reduce the gold logit. This supports the interpretation that the global component is not merely a zero-ablation artifact.

F Additive Decomposition Details

We test whether the mixed-circuit decomposition is approximately additive at the level of causal intervention effects. For each target subtype, we construct disjoint global, family-residual, and subtype-residual components and evaluate interventions that include or exclude each component. We then fit regressions predicting raw margin recovery Δ from component indicators.

Factorial regression. The additive model is

$$\Delta = \beta_0 + \beta_G I_G + \beta_F I_F + \beta_S I_S + \epsilon,$$

where I_G , I_F , and I_S indicate whether the global, family-residual, or subtype-residual component is ablated. Table 14 compares this additive model with a full interaction model.

The additive model captures most of the explainable structure: adding all interaction terms increases R^2 from 0.572 to 0.578, and the predictions of the additive and full-interaction models are nearly identical ($r = 0.995$). Thus, the decomposition should not be read as a literal linear factorization of model internals, but it is a good

Term	Additive	Full interaction
Intercept	−0.254	−0.001
G	2.936	2.492
F	1.009	0.735
S	1.159	0.886
$G:F$	—	0.425
$G:S$	—	0.422
$F:S$	—	0.085
$G:F:S$	—	0.079
R^2	0.572	0.578

Table 14: Factorial regression on raw margin recovery; interaction terms add little beyond the additive model.

functional approximation of how the causal effects combine.

Fixed-budget diagnostic. As a complementary check, we construct an 80-head mixed intervention by combining G_{40} , F_{20} , and S_{20} : the top 40 global heads, 20 family-residual heads excluding those global heads, and 20 subtype-residual heads excluding both previous sets. This keeps the total intervention size fixed at 80 heads.

This fixed-budget result is diagnostic rather than an optimized allocation. Its purpose is to show that family- and subtype-residual components can add

Condition	Recovery	Raw Δ	Clean harm
$G_{40} + F_{20} + S_{20}$	1.550	5.003	−1.012
G_{80} global only	1.464	4.900	−0.990
S_{80} subtype only	1.404	4.715	−1.073
F_{80} family only	1.306	4.254	−1.111
Random 80	−0.091	−0.308	0.523
Same-layer random 80	−0.321	−1.000	0.664

Table 15: Fixed-budget diagnostic (80 heads): the structured mix slightly outperforms single-component circuits; random controls fail.

useful signal beyond the global core under a controlled head budget. Combined with the regression analysis, it supports the mixed-circuit interpretation at the level of causal intervention effects: G is the dominant component, while $F_{\text{fam}(r)}$ and S_r provide smaller positive residual effects.

G Illustrative Gumbel-Sigmoid Intervention

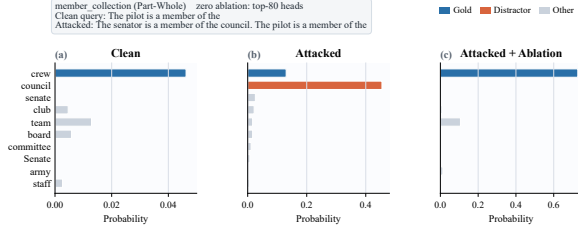


Figure 6: An illustrative factual-distractor example under the ranking-based intervention used in our Gumbel-Sigmoid head search. The clean prompt favors crew; the distractor context shifts probability toward council; and zero-ablating the top-80 heads exported from the learned mask scores reverses this shift.

Figure 6 shows a representative valid attacked case used to visualize the intervention produced by our Gumbel-Sigmoid head search. The figure is illustrative rather than a separate evaluation result; all quantitative analyses in the main text are computed over filtered held-out attack cases.

H Gumbel-Sigmoid Search Details

This appendix provides additional details for the differentiable head-search procedure described in Section 3.4. For each attention head h_i , we introduce a learnable mask logit α_i . During search, we sample a relaxed mask

$$\tilde{m}_i = \sigma \left(\frac{\alpha_i + g_i}{\tau} \right),$$

where g_i is logistic noise induced by the Gumbel reparameterization and τ is a temperature hyperparameter. In the forward pass, we use a straight-through binary mask obtained by thresholding $\tilde{m}_i$, while gradients are passed through the relaxed value. A binary value of $m_i = 1$ means that the corresponding head output is zero-ablated; $m_i = 0$ leaves the head active.

For a batch of attacked prompts $\mathcal{B}$, the search objective is

$$\mathcal{L} = -\mathbb{E}_{x \in \mathcal{B}} \left[M_{\theta}^{\text{abl}}(x) - M_{\theta}(x) \right] + \frac{\lambda}{|\mathcal{H}|} \sum_{i=1}^{|\mathcal{H}|} \sigma(\alpha_i),$$

where $M_{\theta}(x)$ is the attacked margin of the original model, $M_{\theta}^{\text{abl}}(x)$ is the margin under the current differentiable mask, $\mathcal{H}$ is the set of all attention heads, and λ controls sparsity. Since $M_{\theta}(x)$ is constant with respect to the mask for a fixed batch, this objective is equivalent during optimization to maximizing the ablated margin while penalizing the expected number of selected heads.

We anneal the Gumbel-Sigmoid temperature during search and optimize mask logits with gradient descent while keeping all model parameters frozen. After optimization, the learned mask scores define a ranking over heads, and the main experiments use the top $K = 80$ interventions exported from that ranking. We select this value with a held-out K -sweep over $K \in \{20, 40, 60, 80\}$, choosing a setting that provides strong attacked-margin recovery while avoiding excessive clean-prompt harm. For frequency-defined analyses, we additionally report results at $K \in \{20, 40, 80\}$.

For the main LLaMA-3.1-8B experiments, subtype-specific Gumbel searches use 450 optimization steps, while family-shared and global-shared searches use 650 steps because they aggregate more heterogeneous training cases. The seed-stability runs use 300 steps. We use a search batch size of 2, an evaluation batch size of 4, learning rate 0.08, sparsity weight $\lambda = 0.025$, and a Gumbel-Sigmoid temperature annealed from 1.0 to 0.25. The main head budget is $K = 80$, selected from a held-out sweep over $K \in \{20, 40, 60, 80\}$. Each subtype uses 120 train attack pairs and 120 test attack pairs before valid-attack filtering; the main valid-attack threshold is $D \geq 0.5$.

I Artifact use.

We use publicly available model checkpoints and relation-completion resources for research evaluation, following their respective access conditions. Our constructed benchmark consists of synthetic templates and factual lexical-semantic pairs.